

Scenario 1: Student Academic Performance Dashboard

Code :

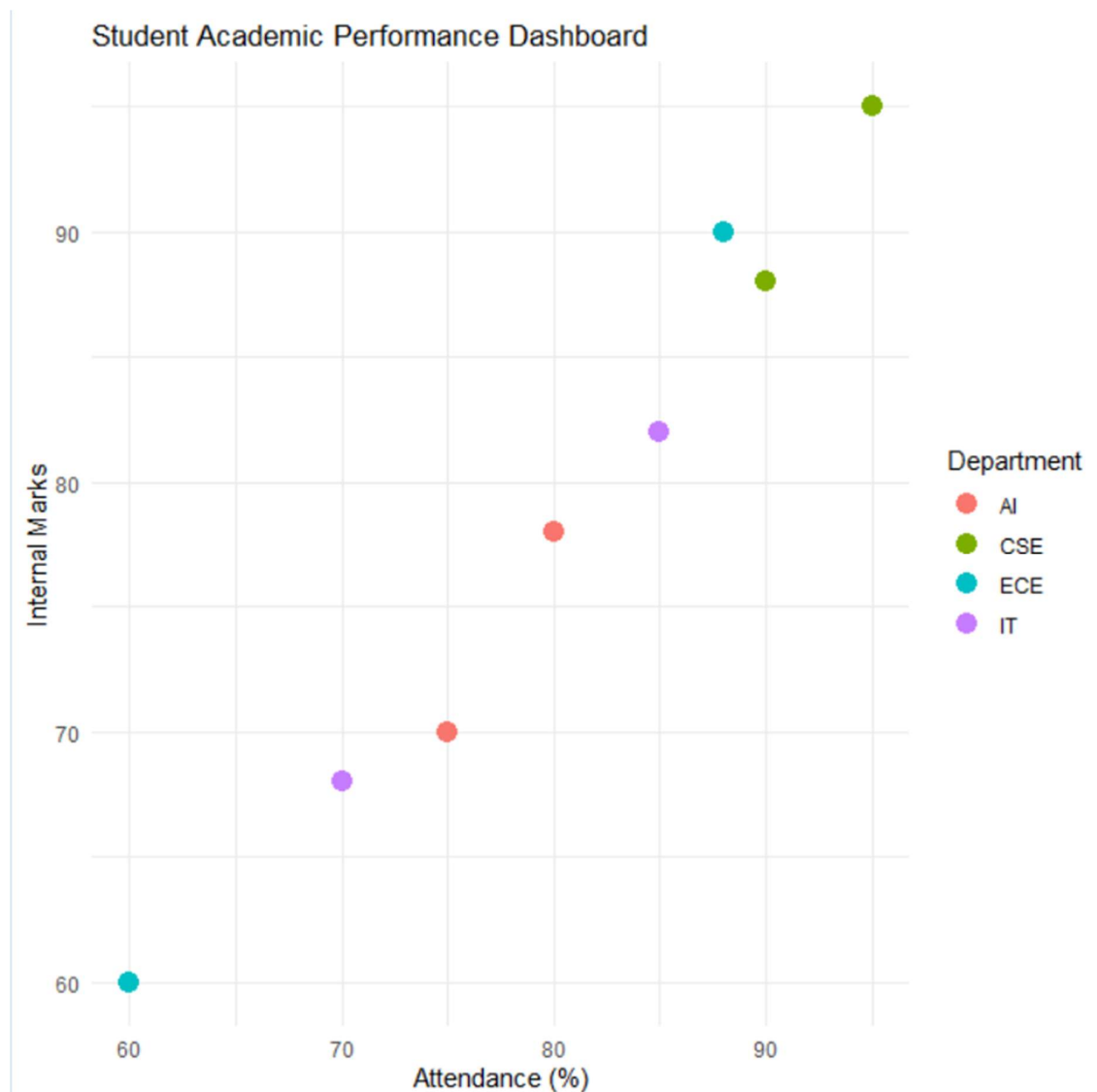
```
# Install ggplot2 if not already installed
# install.packages("ggplot2")

library(ggplot2)

student_data <- data.frame(
  Student_ID = c(101,102,103,104,105,106,107,108),
  Attendance = c(90,75,85,60,95,80,70,88),
  Internal_Marks = c(88,70,82,60,95,78,68,90),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE")
)

# Scatter Plot
ggplot(student_data,
  aes(x = Attendance,
      y = Internal_Marks,
      color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

Output:



Scenario 2: Electric Vehicle Market Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```

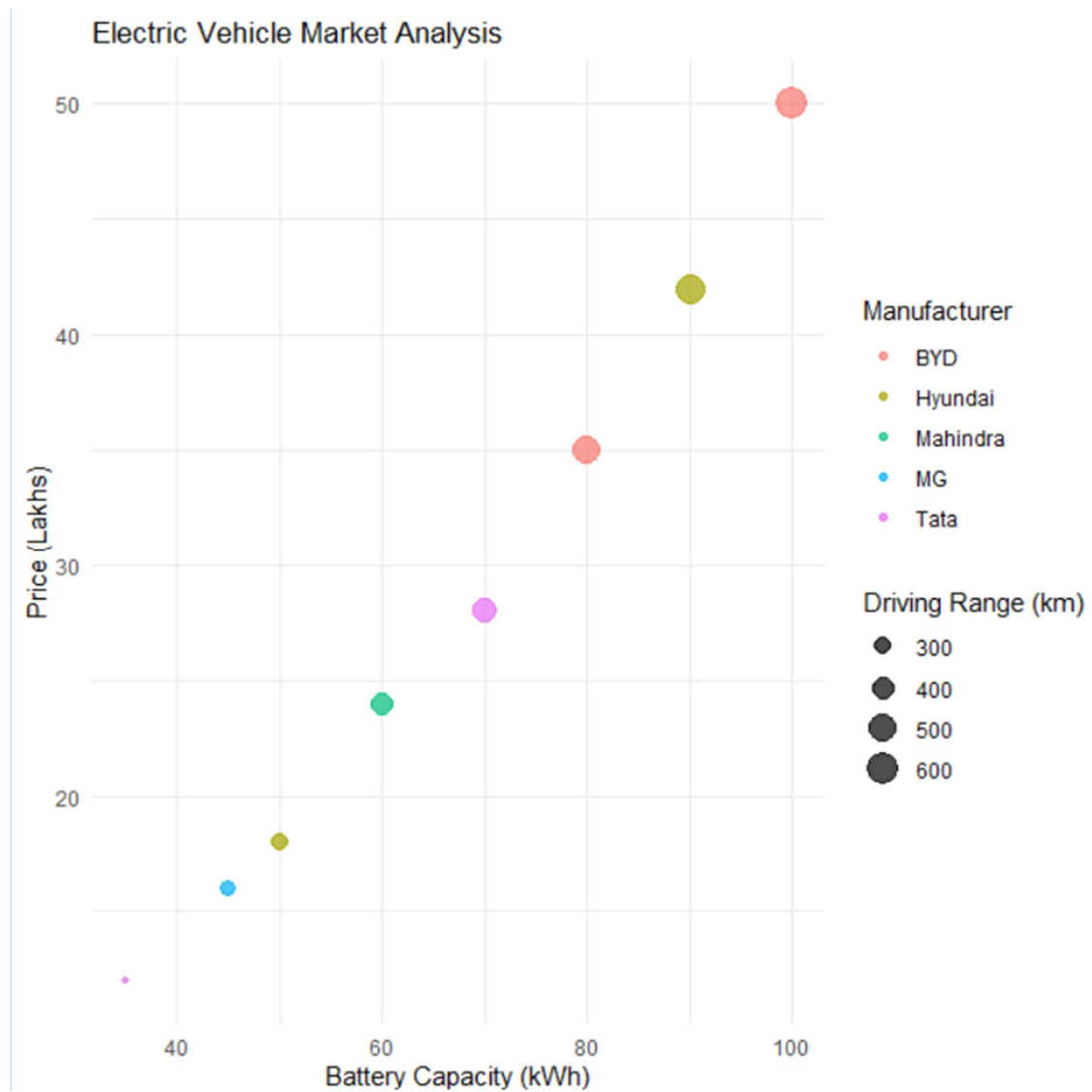
```

ev_data <- data.frame(
  Vehicle_Model = c("Model A", "Model B", "Model C", "Model D",
                    "Model E", "Model F", "Model G", "Model H"),
  Battery_Capacity = c(35, 45, 50, 60, 70, 80, 90, 100),
  Price = c(12, 16, 18, 24, 28, 35, 42, 50), # Price in Lakhs
  Driving_Range = c(220, 280, 300, 380, 420, 500, 550, 620),
  Manufacturer = c("Tata", "MG", "Hyundai", "Mahindra",
                   "Tata", "BYD", "Hyundai", "BYD")
)

# Bubble Chart
ggplot(ev_data,
  aes(x = Battery_Capacity,
      y = Price,
      size = Driving_Range,
      color = Manufacturer)) +
  geom_point(alpha = 0.7) +
  labs(
    title = "Electric Vehicle Market Analysis",
    x = "Battery Capacity (kWh)",
    y = "Price (Lakhs)",
    size = "Driving Range (km)",
    color = "Manufacturer"
  ) +
  theme_minimal()

```

Output:



Scenario 3: Hospital Patient Admission Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```

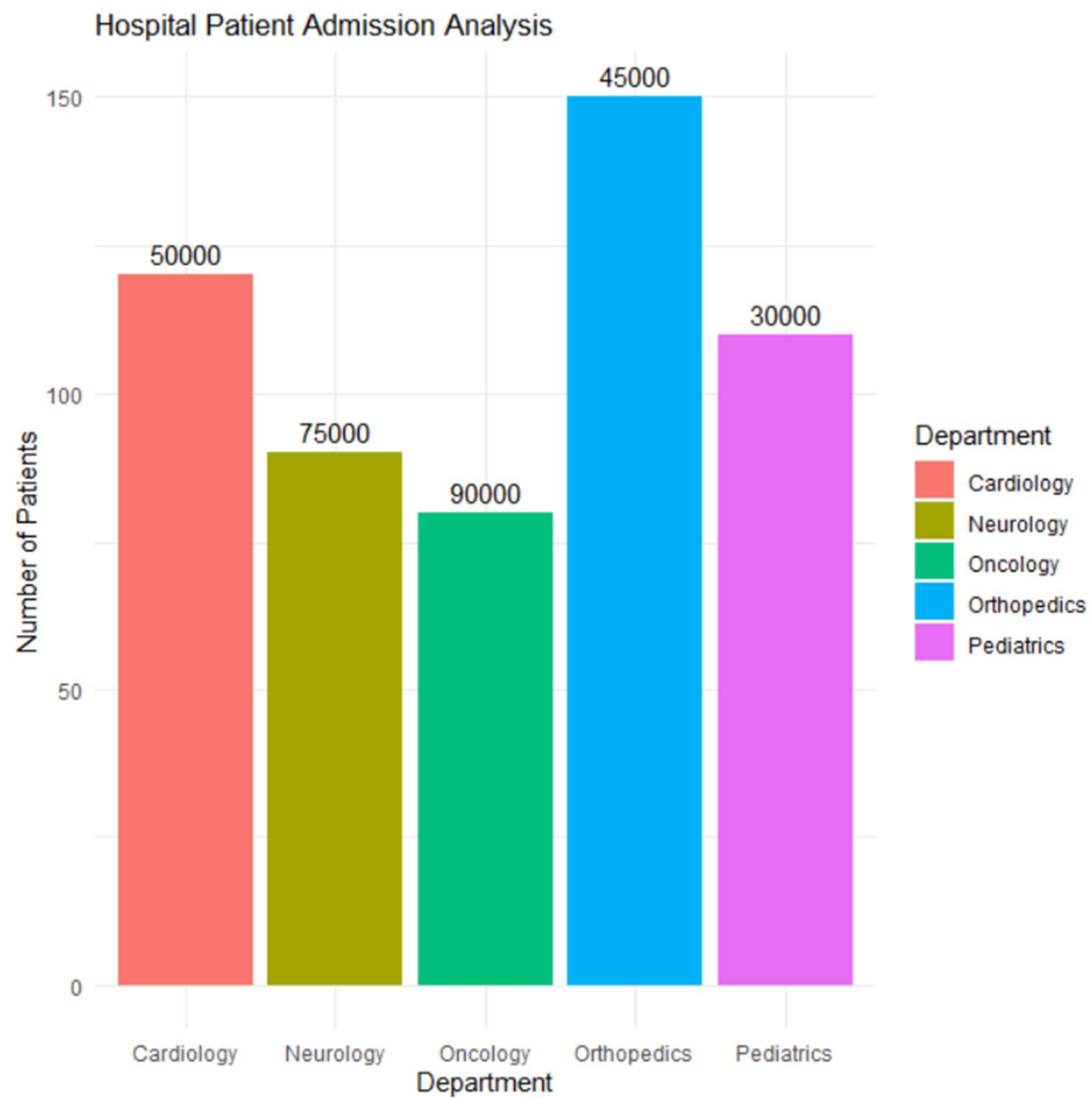
```

hospital_data <- data.frame(
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics", "Oncology"),
  Number_of_Patients = c(120, 90, 150, 110, 80),
  Average_Treatment_Cost = c(50000, 75000, 45000, 30000, 90000)
)

# Bar Chart
ggplot(hospital_data,
  aes(x = Department,
    y = Number_of_Patients,
    fill = Department)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Average_Treatment_Cost),
    vjust = -0.5,
    size = 4) +
  labs(
    title = "Hospital Patient Admission Analysis",
    x = "Department",
    y = "Number of Patients",
    fill = "Department"
  ) +
  theme_minimal()

```

Output:



Scenario 4: Weather Monitoring System

Code:

```
# Install ggplot2 if not installed  
# install.packages("ggplot2")
```

```
library(ggplot2)
```

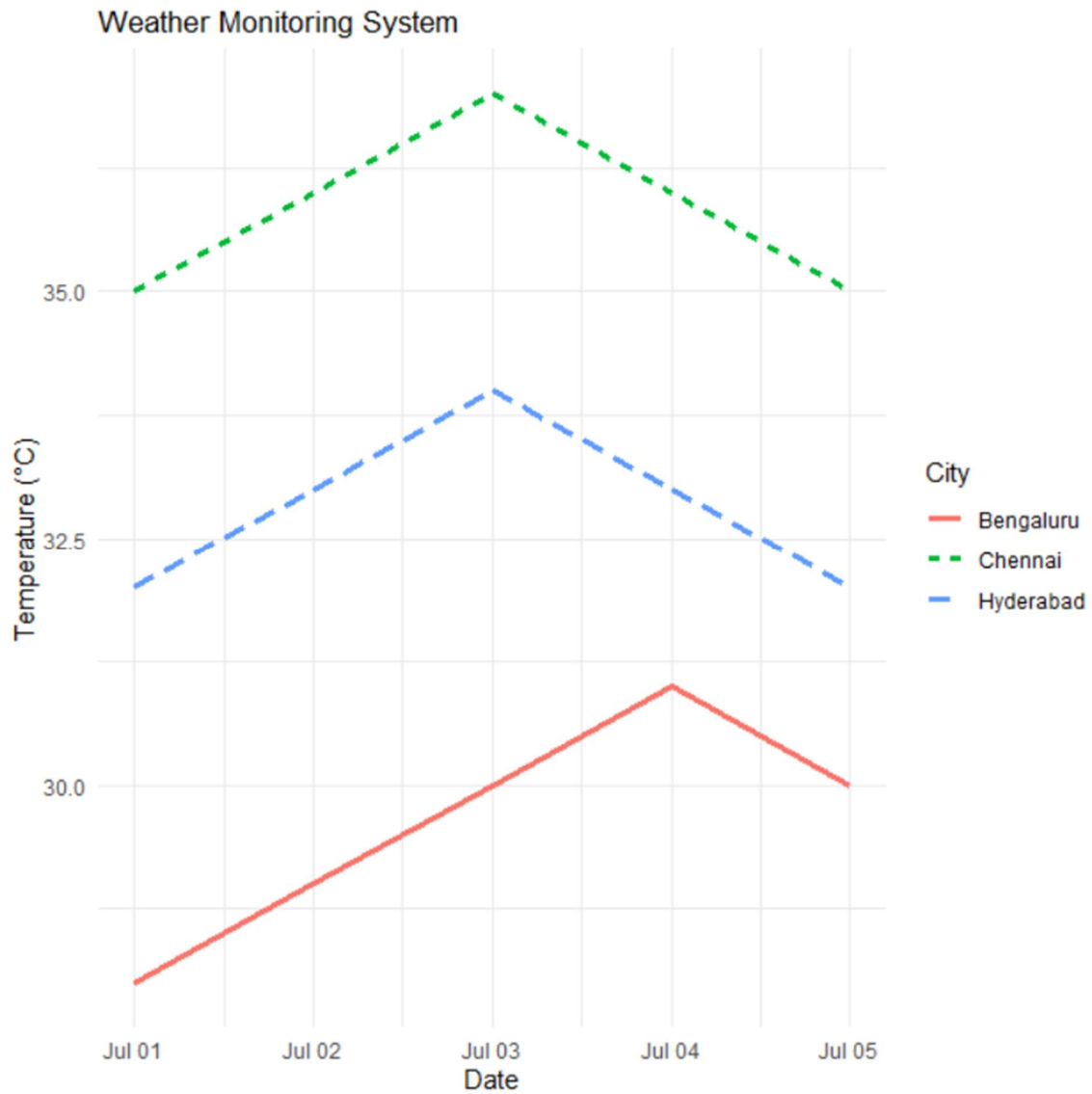
```

# Sample dataset
weather_data <- data.frame(
  Date = rep(as.Date(c("2025-07-01", "2025-07-02", "2025-07-03", "2025-07-04", "2025-07-05")), 3),
  Temperature = c(
    35, 36, 37, 36, 35, # Chennai
    28, 29, 30, 31, 30, # Bengaluru
    32, 33, 34, 33, 32 # Hyderabad
  ),
  City = rep(c("Chennai", "Bengaluru", "Hyderabad"), each = 5)
)

# Line Chart
ggplot(weather_data,
  aes(x = Date,
    y = Temperature,
    color = City,
    linetype = City,
    group = City)) +
  geom_line(size = 1.2) +
  labs(
    title = "Weather Monitoring System",
    x = "Date",
    y = "Temperature (°C)",
    color = "City",
    linetype = "City"
  ) +
  theme_minimal()

```

Output:



Scenario 5: Supermarket Sales Analysis

Code:

```
# Install ggplot2 if not installed
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Sample dataset
```



```

sales_data <- data.frame(
  Month = rep("January", 9),
  Product_Category = rep(c("Groceries", "Beverages", "Snacks"), each = 3),
  Sales_Amount = c(50000, 45000, 48000,
                    30000, 32000, 31000,
                    20000, 22000, 21000),
  Branch = rep(c("Branch A", "Branch B", "Branch C"), 3)
)

# Grouped Bar Chart
ggplot(sales_data,
  aes(x = Product_Category,
      y = Sales_Amount,
      fill = Branch)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Supermarket Sales Analysis",
    x = "Product Category",
    y = "Sales Amount",
    fill = "Branch"
  ) +
  theme_minimal()

```

Output:

